WORKSHEET

Chapter 4: Graphical representation of a sequence and long-term solution of a recurrence relation

1 For each of the following recurrence relations:

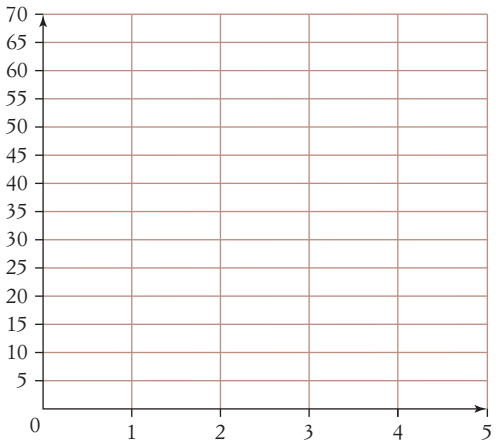
i list the first five terms of the sequence

ii use the grid provided to display the terms of the sequence on a graph.

a $t_{n+1} = t_n - 12, t_n = 70$

i _____, _____, _____, _____, _____

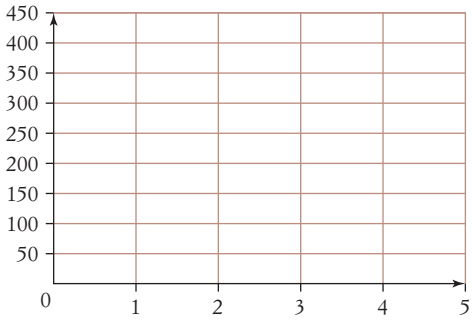
ii



b $t_{n+1} = 3t_n, t_n = 5$

i _____, _____, _____, _____, _____

ii

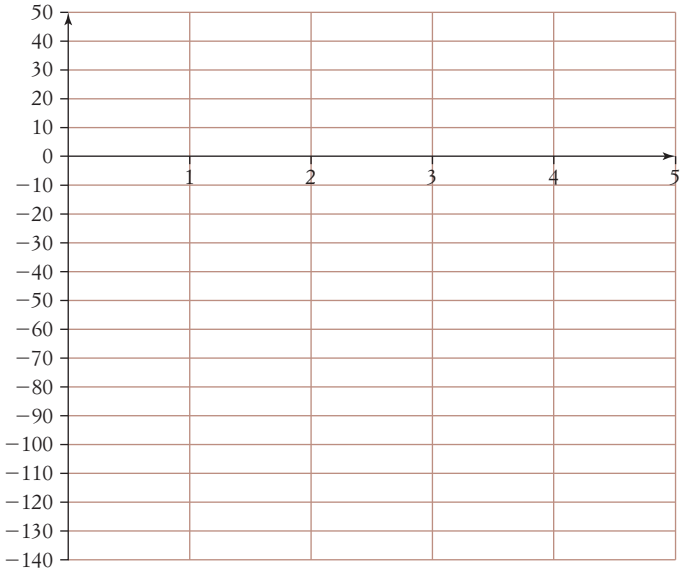


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c $t_{n+1} = -3t_n + 7, t_n = 0$

i _____, _____, _____, _____, _____

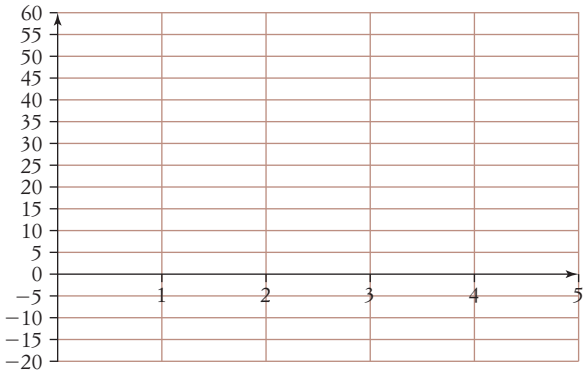
ii



d $t_{n+1} = 0.4t_n - 10, t_1 = 60$

i _____, _____, _____, _____, _____

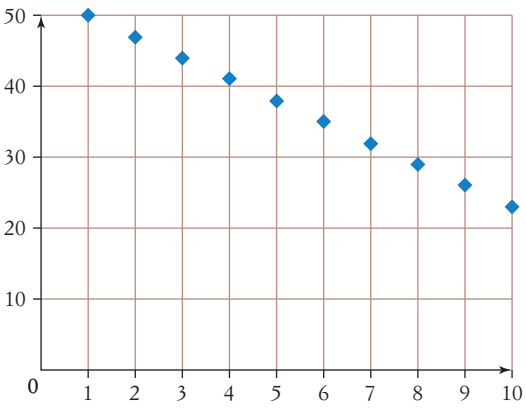
ii



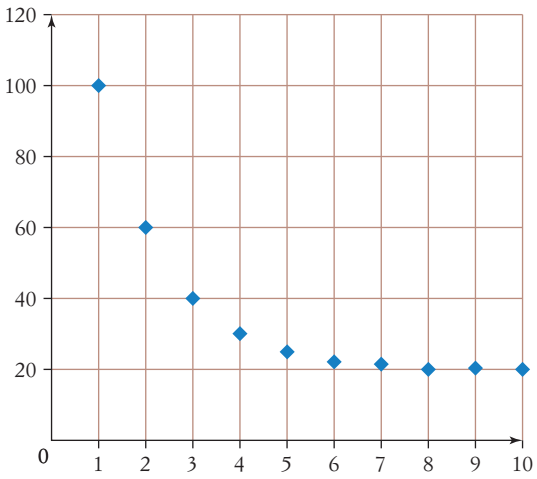
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2 State whether each of the following graphs has a long-term increasing, decreasing or steady-state solution.

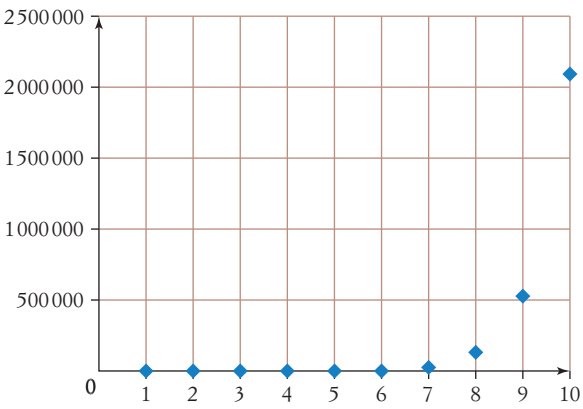
a



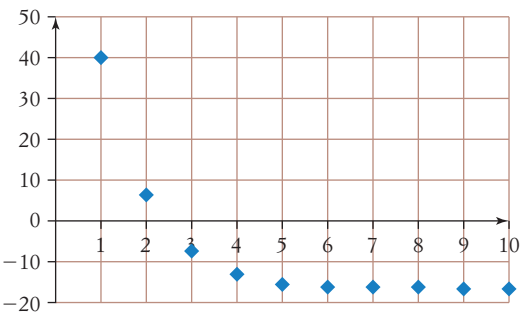
b



c



d



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3 Match each of the following four recurrence relations to the correct graph.

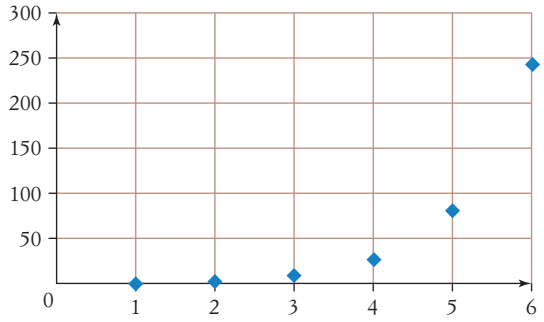
A $t_{n+1} = 0.5t_n + 20, t_n = 7$

B $t_{n+1} = \frac{4}{5}t_n - 8, t_n = 200$

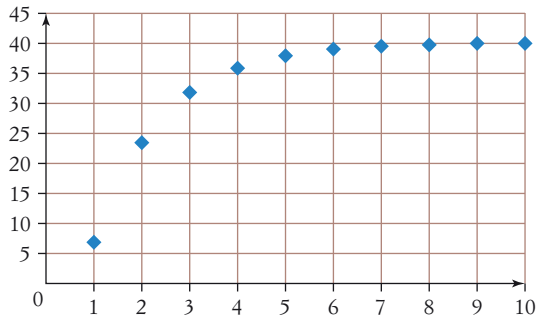
C $t_{n+1} = t_n + 9, t_n = 4$

D $t_{n+1} = 3t_n, t_n = 1$

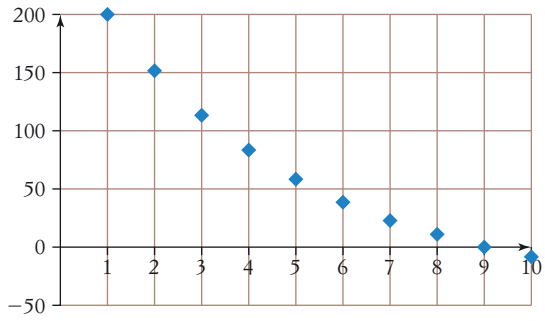
a



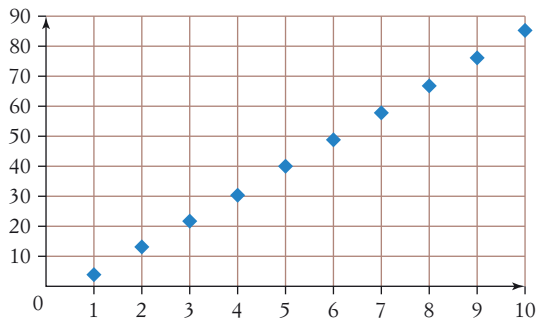
b



c



d



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4 Circle the equations that have a long-term steady-state solution.

a $t_{n+1} = \frac{4}{5}t_n - 8, t_n = 200$

b $t_{n+1} = -3t_n, t_n = 1$

c $t_{n+1} = t_n + 9, t_n = 4$

d $t_{n+1} = 0.5t_n + 20, t_n = 7$

e $t_{n+1} = t_n - 17, t_n = 400$

f $t_{n+1} = 4t_n - 0.5, t_n = 14$

g $t_{n+1} = -\frac{1}{3}t_n, t_n = 16$

h $t_{n+1} = 4t_n - 5, t_n = 1$

i $t_{n+1} = 10 - t_n, t_n = 2$

j $t_{n+1} = 3 - 0.1t_n, t_n = 48$

5 Find the long-term steady state of each of the following recurrence relations. Answer to one decimal place where appropriate.

a $t_{n+1} = 0.5t_n + 20, t_n = 7$

b $t_{n+1} = -\frac{1}{4}t_n + 16, t_n = 10$

c $t_{n+1} = \frac{1}{2}t_n, t_n = 200$

d $t_{n+1} = -0.9t_n + 1, t_n = 10$

6 The wild pig population on Banas Island in 2013 was 2600. Each year, the population increases by 33%, and rangers have to cull 900 pigs to keep the population under control. What will be the long-term size of the wild pig population? Answer to the nearest whole number.

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7 Merida inherits $2 000 000 which she invests. Each year, she uses 10% of the investment for living expenses, and the value of the investment increases by $142 000. What is the long-term value of Merida's investment?

8 As a result of a summer of heavy rains, red algae developed on Lake Tuka. Once established, this algae is impossible to eradicate; however, if treated once a year, it can be maintained at a low level. Before treatment began, the estimated quantity of algae on the lake was 5.7 tonnes. Each year, the treatment reduces the amount of algae by 81%, but the algae regenerates by 0.42 tonnes. In the long term, how much red algae will there be in the lake? Answer to the nearest kilogram.

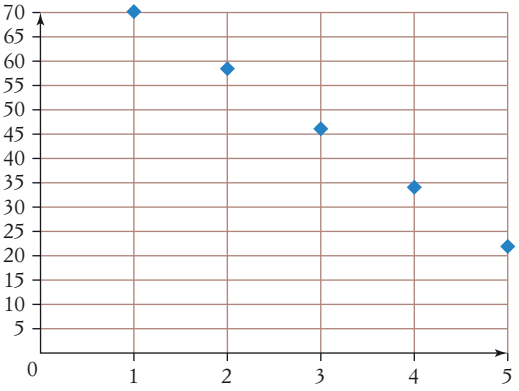
9 Boiling water is poured into a tank where the temperature drops by 46% every hour. Each hour, the water temperature is increased by 12° using a gas heater. What will the water temperature be once it reaches a steady state? Answer to one decimal place.

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Answers

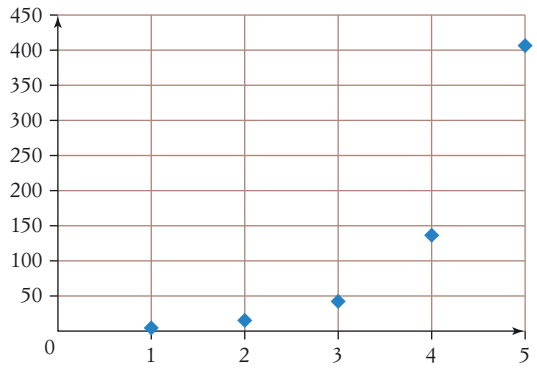
1 a i 70, 58, 46, 34, 22

ii



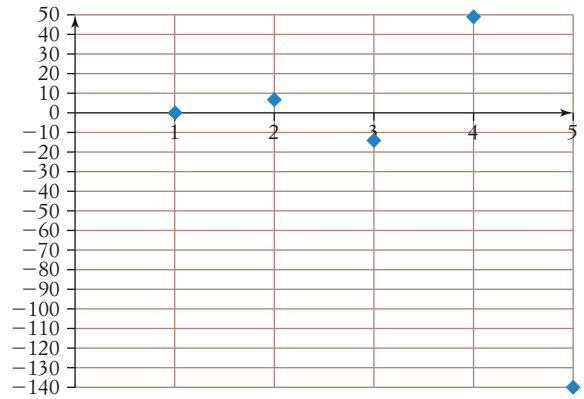
b i 5, 15, 45, 135, 405

ii



c i 0, 7, −14, 49, −140

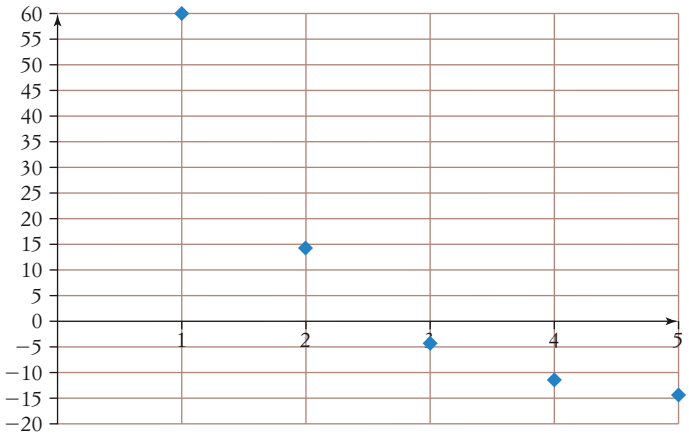
ii



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d **i** 60, 14, −4.4, −11.76, −14.704

ii



2 **a** Decreasing

b Long-term steady state

c Increasing

d Long-term steady state

3 **a** D

b A

c B

d C

4 a, d, g, j

5 **a** 40

b 12.8

c 0

d 0.5

6 2727

7 $1 420 000

8 519 kg

9 26.1°

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